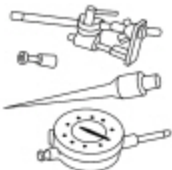


Forward Clutch Assembly

 [Ford Ecat Electronic Parts Catalog](#)

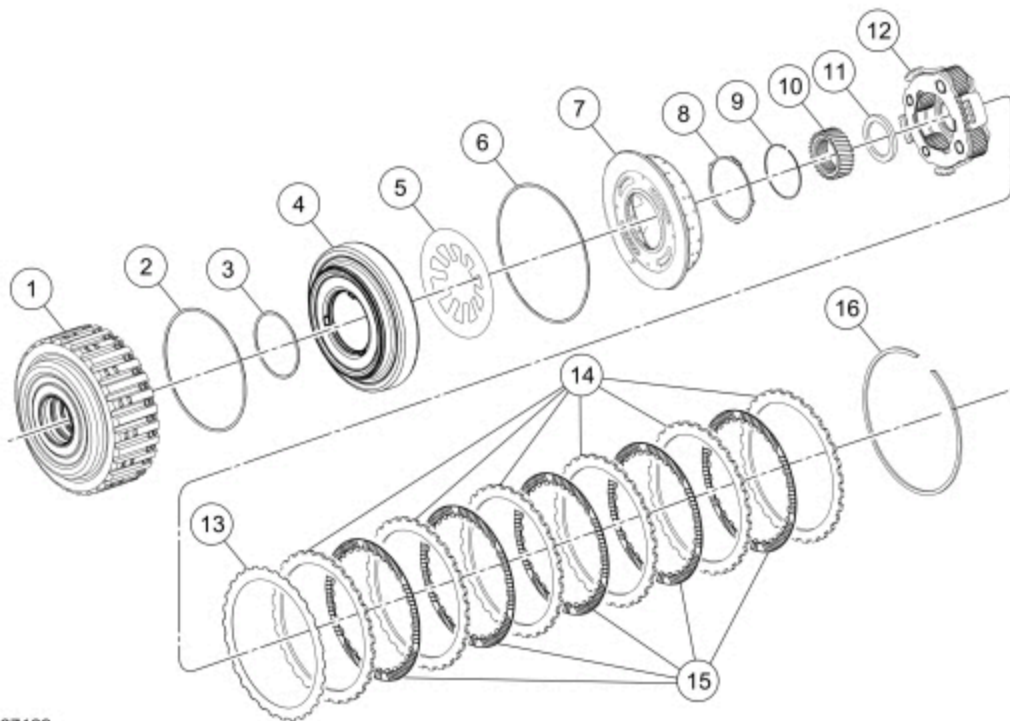
Special Tool(s)

 ST2881-A	Compressor, Clutch Spring 307-525
 ST1214-A	Dial Indicator Gauge with Holding Fixture 100-002 (TOOL-4201-C) or equivalent
 ST2892-A	End Play Gauge, Clutch 307-555

Material

Item	Specification
Motorcraft® MERCON® LV Automatic Transmission Fluid XT-10-QLVC (US); CXT-10-LV12 (Canada)	MERCON® LV

Forward Clutch Assembly (A Clutch)

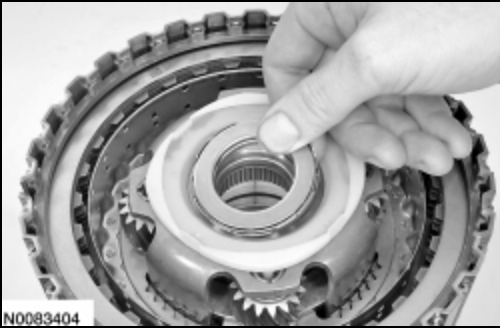


N0167189

Item	Part Number	Description
1	7A360 7A360	Forward clutch drum
2	7A548 7A548	Forward clutch piston outer O-ring seal
3	7A548 7A548	Forward clutch piston inner O-ring seal
4	7A262 7A262	Forward clutch piston
5	7A480 7A480	Forward clutch piston return spring
6	7A548 7A548	Forward clutch balance dam O-ring seal
7	7H360 7H360	Forward clutch balance dam
8	7H365 7H365	Forward clutch balance dam retaining ring
9	7H579 7H579	Sun gear retaining ring
10	7D063 7D063	Sun gear
11	7H375 7H375	Bearing (T1)
12	7A398 7A398	Planetary gear
13	7E085 7E085	Forward clutch cushion spring
14	7E314 7E314	Forward clutch external splined steel plates (quantity model dependent)
15	7B164 7B164	Forward clutch internal splined friction plates (quantity model dependent)
16	7D483 7D483	Forward clutch plate retaining ring

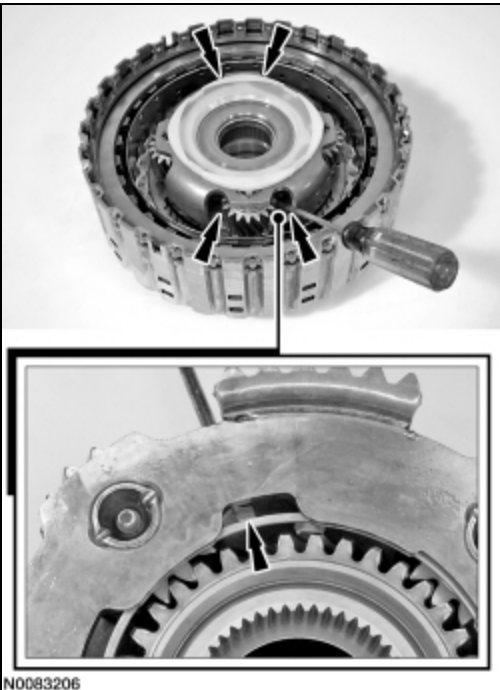
Disassembly

1. If the T2 thrust bearing is on the forward clutch assembly, remove it from the forward clutch assembly.



2. Remove the front planetary carrier and sun gear from the forward clutch assembly.

- With a small screwdriver or a suitable pick inserted into the front planetary assembly, push on the retaining ring in 4 places while lifting up on the front planetary assembly.



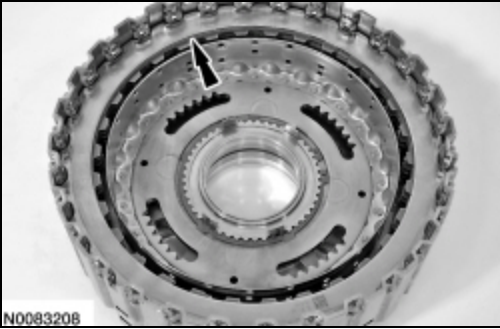
3. Remove the forward clutch pack snap ring.



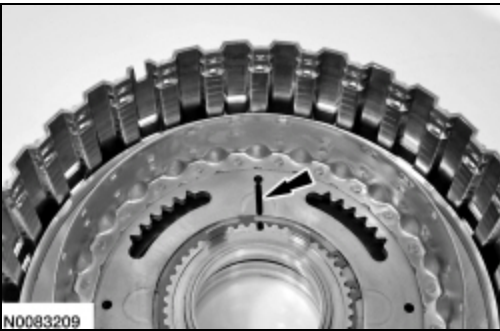
4. **NOTE:** *Inspect the forward clutch drum and the friction and steel plates for damage. Install new components as necessary.*

Remove and inspect the forward clutch steel and friction plates.

- If the plates are damaged, install a new forward clutch assembly.
- If no damage is indicated, the forward clutch may be reassembled.

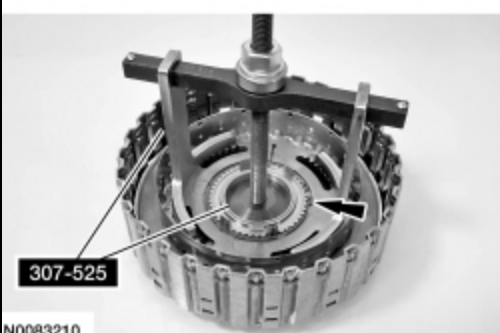
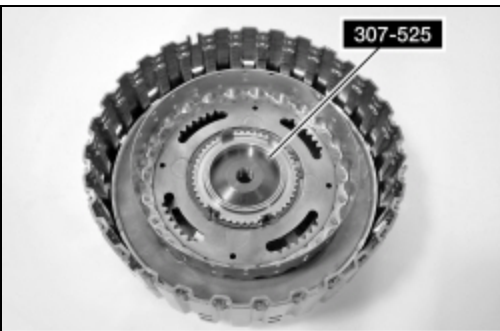


5. For correct alignment during assembly, make an identifying mark on the balance dam and the center part of the housing.

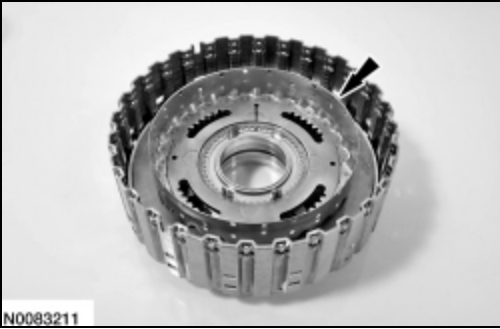


6. **NOTE:** *The bottom of the Clutch Spring Compressor will need to be installed from the back side of the forward clutch assembly.*

Using the Clutch Spring Compressor, slightly collapse the balance dam to gain access to the retaining ring and remove the retaining ring.



7. Remove the balance dam.



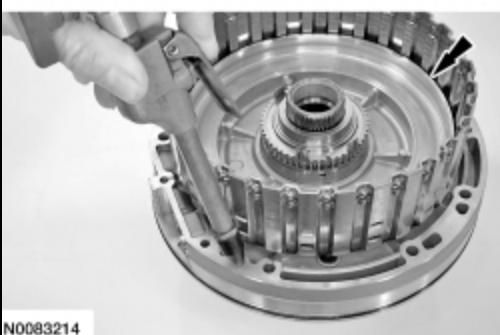
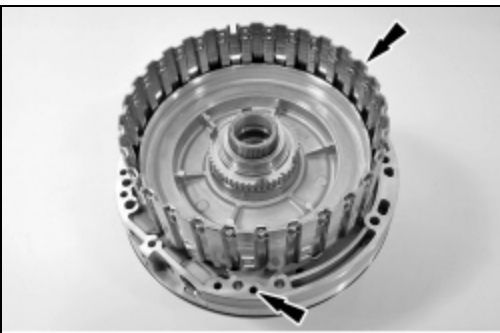
8. Remove and discard the O-ring seal from the balance dam.



9. Remove the forward clutch return spring.



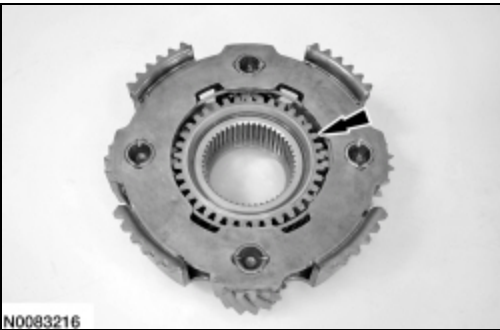
10. Install the forward clutch on the front pump to remove the forward clutch piston. Apply a small amount of shop air to remove the piston.



11. Remove and discard the inner and outer piston O-ring seals.

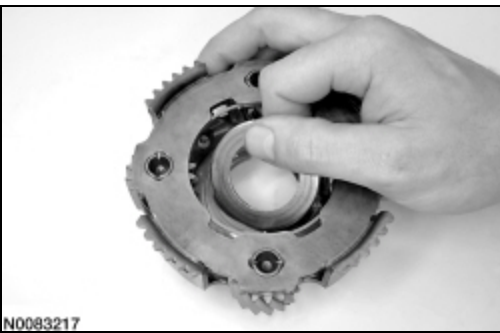


12. Remove the sun gear from the planet assembly.

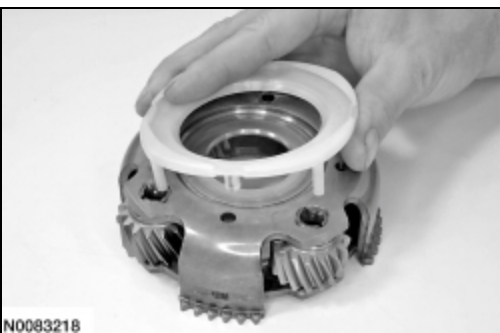


13. **NOTE:** *The T1 thrust bearing may have stuck to the sun gear when it was removed.*

Remove the T1 bearing from the planetary carrier.



14. Remove the transmission fluid collector plate from the front planetary carrier.



15. Clean and inspect the components of the forward clutch assembly for damage and install new components as necessary.

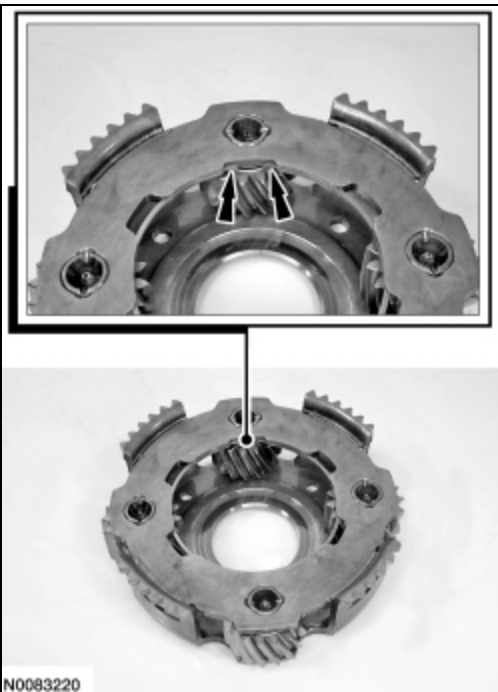
Assembly

1. **NOTE:** *If a new planetary assembly is being installed, the new planetary assembly may not come with the snap ring installed.*

Install the planetary assembly snap ring, if necessary.



2. Make sure the planetary snap ring is oriented as shown.



3. Install the transmission fluid collector plate on the front planetary carrier.



4. Install the T1 bearing onto the sun gear.



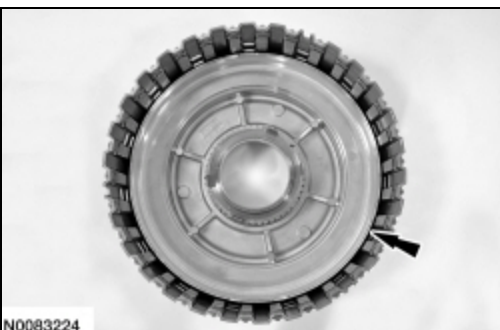
5. Install the sun gear into the planet assembly with the T1 bearing facing down and the recessed part of the gear up.



6. Install new O-ring seals onto the forward clutch piston.



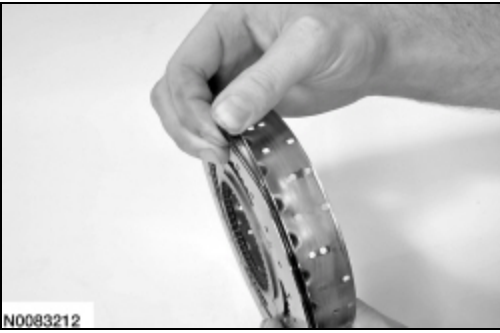
7. Lightly coat the O-ring seals with clean transmission fluid and install the forward clutch piston into the forward clutch drum.



8. Install the forward clutch return spring.



9. Install a new O-ring seal onto the forward clutch balance dam. Lightly coat the O-ring seal with clean transmission fluid.

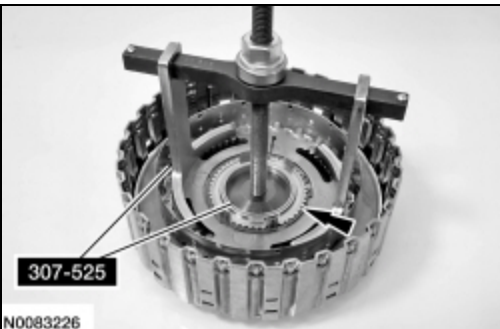


10. Align the marks made during disassembly and install the balance dam into the forward clutch drum.



11. **NOTE:** *Prior to releasing the Clutch Spring Compressor, make sure that the snap ring is fully seated.*

Using the Clutch Spring Compressor, slightly collapse the balance dam to install the retaining ring. Install the retaining ring.

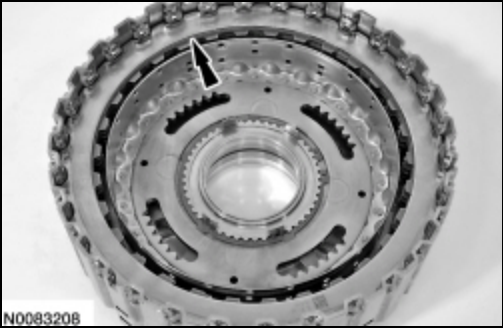


12. **NOTE:** *Friction and steel plate quantity will vary based on engine displacement. For additional information, refer to the Clutch Plate Quantity chart in the Specifications portion of this section.*

Soak the new forward clutch plates in clean transmission fluid.



13. Install the new forward clutch plates, starting with the wave spring then a steel plate and alternating between friction and steel plates.

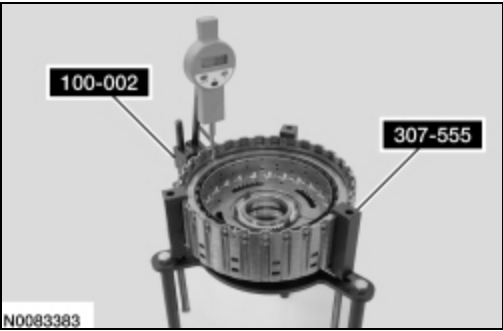


14. Install the forward clutch plate snap ring.

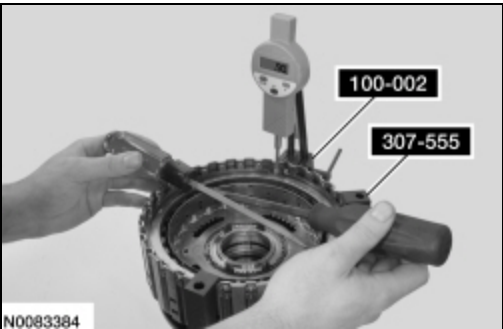


15. **NOTE:** All forward clutch plates, friction and steel are of a wave-type design.

Install the forward clutch assembly into the Clutch End Play Gauge and install the Dial Indicator Gauge with Holding Fixture and position the plunger so it fits into the opening of the snap ring.



16. With the Dial Indicator Gauge set at zero, lift up on the steel pressure plate so it is against the select fit snap ring. Record this reading as reading A.



17. Rotate the forward clutch assembly 180 degrees from the opening of the snap ring, take a second reading and record this reading as reading B.

Description	Reading
Reading A	
Reading B	

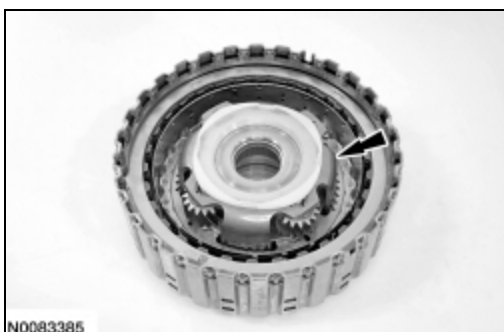
Description	Reading
Add reading A to reading B for a total end clearance	
Divide the total reading by 2 for an average end clearance	

18. **NOTE:** *If the final measurement is not within specification, install a new snap ring until the correct specification is achieved.*

- 0.71-1.10 mm (0.028-0.043 in)

If the free pack end clearance between the bottom of the snap ring and the top of the pressure plate is high, remeasure using a thicker select fit snap ring. If the free pack end clearance between the bottom of the snap ring and the top of the pressure plate is low, remeasure using a thinner select fit snap ring.

19. Install the sun gear and planetary gearset into the forward clutch drum. Make sure it snaps and locks in place.



20. Install the T2 thrust bearing.

